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Multipurpose module, module fixing device, and locking assembly for vehicle

Abstract

A vehicle includes a multipurpose module replaceably coupled to a luggage zone of the vehicle, and a module fixing device coupled to the multipurpose module and configured to fix the multipurpose module to a vehicle body of the vehicle, in which the module fixing device includes one or more locking assemblies configured to fix the multipurpose module by being matched and inserted into a fixing hole provided in the multipurpose module, and a locking controller configured to control the locking assembly, in which the locking assembly includes a shape memory spring made of a shape memory alloy, a fixing protrusion connected to the shape memory spring and configured to move forward to be inserted into the fixing hole or move rearward to be separated from the fixing hole in accordance with extension or contraction of the shape memory spring, and a heater configured to apply heat to the shape memory spring, and in which the shape memory spring operates the fixing protrusion while being extended or contracted by the applied heat.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application claims the benefit of priority to Korean Patent Application No. 10-2022-0069754, filed on Jun. 8, 2022, in the Korean Intellectual Property Office, the disclosure of which is incorporated herein by reference.

TECHNICAL FIELD

(2) The disclosed disclosure relates to a multipurpose module, a module fixing device, and a locking assembly for a vehicle.

BACKGROUND

(3) To preoccupy future mobility markets and technologies, worldwide vehicle manufacturers discover, introduce, and advertise various concepts based on electric vehicles. Among the various concepts, the representative concept is a purpose-built vehicle (PBV). The PBV may be used for passenger transportation, a mobile office, courier service, cargo transportation, a game room, a cafeteria, a shop, and the like by variously arranging seats. The PBV provides concepts that may be used for various purposes by easily changing structures according to various purposes instead of a single purpose.

(4) To implement the various concepts, a vehicle is divided into a body and a platform, and a body region is configured to conform to the purpose and coupled to the platform, which may provide a vehicle with various concepts suitable for the purpose while replacing several bodies in respect to a single platform.
(5) However, a separate facility is required to separate and couple the body and the platform, and spaces and workers are required to replace the bodies and store and manage the bodies that are not used. For this reason, it is difficult to obtain an advantage over vehicles in the related art.

SUMMARY

(6) In the case of the concept of the purpose-built vehicle (PBV) in which the body and the platform are separated from or coupled to each other, the body region is divided into a drive zone essential in common to drive the vehicle, and a luggage zone, i.e., a region that may be changed depending on the purpose of the vehicle. The concept for replacing the entire body including both the drive zone and the luggage zone requires a separate facility and workers required to manage the vehicle and spaces for storing and managing the replaced body. For this reason, it is difficult to obtain an advantage over vehicles in the related art in terms of management for each vehicle. According to one aspect of the disclosure proposed to cope with inefficient parts of the concept while maintaining the purpose of the purpose-built vehicle that is the PBV concept in view of each vehicle, a body and a platform of the PBV are coupled to each other, and a body region includes a drive zone, which is essential in common, and a luggage zone that may be changed depending on the purpose. In this case, the drive zone is fixed in common to drive the vehicle. The luggage zone provides various modules that may be changed depending on the purpose of the vehicle and used for passenger transportation, a mobile office, courier service, cargo transportation, a game room, a cafeteria, a shop, and the like by variously arranging seats. The luggage zone provides a structure that may be separated from or coupled to the body. An aspect of the disclosure is to provide a multipurpose module, module fixing device, and locking assembly for a vehicle, in which the locking assembly is included in a luggage zone and operated by a shape memory spring, and the multipurpose module is coupled to the module fixing device equipped with a constituent component suitable for the purpose, such that the purpose of the vehicle may be easily changed.
(7) In accordance with an aspect of the disclosure, a vehicle includes a multipurpose module replaceably coupled to a luggage zone of the vehicle, and a module fixing device coupled to the multipurpose module and configured to fix the multipurpose module to a vehicle body of the vehicle, in which the module fixing device includes one or more locking assemblies configured to fix the multipurpose module by being matched and inserted into a fixing hole provided in the multipurpose module, and a locking controller configured to control an operation of the locking assembly, in which the locking assembly includes a shape memory spring made of a shape memory alloy, a fixing protrusion connected to the shape memory spring and configured to operate forward to be inserted into the fixing hole or operate rearward to be separated from the fixing hole in accordance with extension or contraction of the shape memory spring, and a heater configured to apply heat to the shape memory spring, and in which the shape memory spring operates the fixing protrusion while being extended or contracted by the applied heat.
(8) The fixing protrusion may operate forward when no heat is applied to the shape memory spring, and the fixing protrusion may operate rearward when heat is applied to the shape memory spring.
(9) The locking assembly may further include a reaction force spring configured to apply an elastic force to the fixing protrusion in a forward direction, and driving power implemented by shape deformation of the shape memory spring may be higher than the elastic force of the reaction force spring.
(10) A rail may be provided on the multipurpose module, and a guide may be provided on the module fixing device, such that the module fixing device and the multipurpose module may be coupled as the rail is inserted into the guide, and the locking assembly may be provided in the guide and matched and inserted into the fixing hole provided in the rail.

(11) The locking assembly may be provided at one side of the guide and protrude when the fixing protrusion operates forward, a protrusion fixing groove may be provided at the other side of the guide, and the fixing protrusion may be inserted into the protrusion fixing groove when the fixing protrusion operates forward.
(12) The multipurpose module may include a fixing base on which a constituent component is mounted, and the rail may be provided at a lower side of the fixing base.
(13) The rail may be a T-shaped rail including a web coupled to the fixing base, and a flange coupled to the web, and the fixing hole may be provided in the web.
(14) The multipurpose module may be coupled to the module fixing device in a forward/rearward direction.
(15) In accordance with another aspect of the disclosure, a module fixing device for a vehicle includes one or more locking assemblies configured to fix a multipurpose module by being matched and inserted into a fixing hole provided in the multipurpose module replaceably coupled to a luggage zone of a vehicle, and a locking controller configured to control an operation of the locking assembly, in which the locking assembly includes a shape memory spring made of a shape memory alloy, a fixing protrusion connected to the shape memory spring and configured to operate forward to be inserted into the fixing hole or operate rearward to be separated from the fixing hole in accordance with extension or contraction of the shape memory spring, and a heater configured to apply heat to the shape memory spring, and in which the shape memory spring operates the fixing protrusion while being extended or contracted by the applied heat.
(16) The fixing protrusion may operate forward when no heat is applied to the shape memory spring, and the fixing protrusion may operate rearward when heat is applied to the shape memory spring.
(17) The locking assembly may further include a reaction force spring configured to apply an elastic force to the fixing protrusion in a forward direction, and driving power implemented by shape deformation of the shape memory spring may be higher than the elastic force of the reaction force spring.
(18) The module fixing device may have a guide into which a rail provided on the multipurpose module is inserted, such that the module fixing device and the multipurpose module may be coupled as the rail is inserted into the guide, and the locking assembly may be provided in the guide and matched and inserted into the fixing hole provided in the rail.
(19) The locking assembly may be provided at one side of the guide and protrude when the fixing protrusion operates forward, a protrusion fixing groove may be provided at the other side of the guide, and the fixing protrusion may be inserted into the protrusion fixing groove when the fixing protrusion operates forward.
(20) In accordance with another aspect of the disclosure, a multipurpose module is replaceably coupled to a vehicle and includes a fixing hole into which a locking assembly provided and operated in the vehicle is matched and inserted, and a rail inserted into a guide provided in the vehicle and configured to couple the multipurpose module to the vehicle.
(21) The fixing hole may be provided in the rail.
(22) The multipurpose module may further include a fixing base on which a constituent component is mounted, and the rail may be provided at a lower side of the fixing base.
(23) The rail may be a T-shaped rail including a web coupled to the fixing base, and a flange coupled to the web, and the fixing hole may be provided in the web.
(24) In accordance with another aspect of the disclosure, a locking assembly includes a shape memory spring made of a shape memory alloy, a fixing protrusion connected to the shape memory spring and configured to operate forward to be inserted into a fixing hole provided in a coupling object or operate rearward to be separated from the fixing hole in accordance with extension or contraction of the shape memory spring, and a heater configured to apply heat to the shape memory spring, and in which the shape memory spring operates the fixing protrusion while being extended

or contracted by the applied heat.
(25) The fixing protrusion may operate forward when no heat is applied to the shape memory spring, and the fixing protrusion may operate rearward when heat is applied to the shape memory spring.
(26) The locking assembly may further include a reaction force spring configured to apply an elastic force to the fixing protrusion in a forward direction, in which driving power implemented by shape deformation of the shape memory spring is higher than the elastic force of the reaction force spring.

Description

BRIEF DESCRIPTION OF THE FIGURES
(1) These and/or other aspects of the disclosure will become apparent and more readily appreciated from the following description of the embodiments, taken in conjunction with the accompanying drawings of which:
(2) FIG. **1** is a view illustrating an example of a vehicle according to the embodiment;
(3) FIG. **2** is a view schematically illustrating a module fixing device of the vehicle according to the embodiment;
(4) FIG. **3** is a view schematically illustrating a multipurpose module according to the embodiment;
(5) FIG. **4** is a view schematically illustrating a multipurpose module according to another embodiment;
(6) FIG. **5** is a front view of a module fixing device according to the embodiment;
(7) FIG. **6** is a top plan view of the module fixing device according to the embodiment;
(8) FIGS. **7** and **8** are cross-sectional front views of the module fixing device according to the embodiment;
(9) FIG. **9** is a view schematically illustrating an operation of the module fixing device according to the embodiment;
(10) FIGS. **10** and **11** are views schematically illustrating that the multipurpose module is coupled to the vehicle according to the embodiment;
(11) FIGS. **12** and **13** are views schematically illustrating an operation of the module fixing device according to the embodiment; and
(12) FIGS. **14**A, **14**B, **14**C, **14**D, and **14**E are views schematically illustrating multipurpose module according to the embodiment.
DETAILED DESCRIPTION
(13) Like reference numerals indicate like constituent elements throughout the specification. The present specification does not explain all the elements in the embodiments, and the general contents in the technical field to which the present disclosure pertains or the contents repeatedly described in the embodiments will be omitted.
(14) Throughout the present specification, when one constituent element is referred to as being “connected to” another constituent element, one constituent element can be “directly connected to” the other constituent element, and one constituent element can also be “indirectly connected to” the other constituent element. The indirect connection includes a connection through a wireless communication network.
(15) In addition, unless explicitly described to the contrary, the word “comprise/include” and variations such as “comprises/includes” or “comprising/including” will be understood to imply the inclusion of stated elements, not the exclusion of any other elements.
(16) Singular expressions include plural expressions unless there is an exception in the context.
(17) In addition, the term “part,” “device,” “block,” “member,” “module,” or the like can mean a unit that processes one or more functions or operations. For example, the term may mean at least

one process processed by at least one hardware stored in a field-programmable gate array (FPGA), an application specific integrated circuit (ASIC), or the like, or at least one software or processor stored in a memory.
(18) The reference numeral assigned to each step is used to identify each step, and the reference numeral does not represent the order between the steps. Each step may be performed regardless of the specified order unless the specified order is clearly described in the context.
(19) Hereinafter, embodiments of a vehicle and a method of controlling the same according to one aspect will be described in detail with reference to the accompanying drawings.
(20) FIG. **1** is a view illustrating an example of a vehicle according to the embodiment.
(21) Referring to FIG. **1**, a vehicle **10** according to an embodiment of the present disclosure includes a multipurpose module **200** replaceably coupled to a luggage zone **11** of the vehicle **10**, and a module fixing device **100** coupled to the multipurpose module **200** and configured to fix the multipurpose module **200** to a vehicle body of the vehicle **10**.
(22) The vehicle **10** illustrated in FIG. **1** may include road wheels W used to allow the vehicle **10** to travel, a driving zone used to drive the vehicle **10**, and the luggage zone **11** used for various purposes.
(23) Various constituent components illustrated in FIG. **14** may be mounted in the luggage zone **11** and perform functions suitable for the purpose of the vehicle. For example, the luggage zone **11** may be equipped with a luggage loading storage for transporting large-scale freight in FIG. **14**A, a freezer and refrigerator for transporting frozen items in FIG. **14**B, module for courier freight transport in FIG. **14**C, seats for transporting passengers in FIG. **14**D, or a cooking facility used for a food truck in FIG. **14**E. In the present disclosure, the multipurpose module **200** equipped with various constituent components is coupled to the luggage zone **11** of the vehicle **10**, such that the constituent components may be easily changed.
(24) The multipurpose module **200** may include the constituent component suitable for the purpose of the vehicle **10** so that the vehicle **10** may be used for various purpose. Therefore, the constituent components of the vehicle **10** may be temporarily changed by changing the multipurpose module **200** coupled to the luggage zone **11** of the vehicle **10**, which makes it possible to use the vehicle **10** for various purposes.
(25) The configuration in which the multipurpose module **200** coupled to the luggage zone **11** in the body of the vehicle **10** is changed is advantageous in changing modules and storing and managing the modules because the size of the module is smaller than the body in comparison with a configuration in which a platform and the body of the vehicle **10** are separated and the body is changed to constitute the vehicle.
(26) In the embodiment of the present disclosure, the module fixing device **100** and the multipurpose module **200** may be coupled to each other in a sliding manner.
(27) In the embodiment of the present disclosure, the multipurpose module **200** may be coupled to the module fixing device **100** in a forward/rearward direction. That is, the multipurpose module **200** may be coupled to the vehicle **10** while sliding in a direction from the rear side to the front side of the vehicle **10**. To this end, the multipurpose module **200** has rails **220**, the module fixing device **100** has guides **130** into which the rails **220** are inserted, and the rails **220** are inserted into the guides **130**, such that the module fixing device **100** and the multipurpose module **200** may be coupled to each other.
(28) Hereinafter, detailed structures of the module fixing device **100** and the multipurpose module **200** will be described.
(29) FIG. **2** is a view schematically illustrating a module fixing device of the vehicle according to the embodiment.
(30) Referring to FIG. **2**, the module fixing device **100** according to the embodiment of the present disclosure may include one or more locking assemblies **110** configured to fix the multipurpose module **200** by being matched with and inserted into fixing holes **210** provided in the multipurpose

module **200**, and a locking controller **120** configured to control an operation of the locking assembly **110**.
(31) As illustrated in FIG. **2**, the module fixing device **100** may include a plurality of locking assemblies **110**. The locking controller **120** configured to control the operation of the locking assembly **110** may allow the locking assembly **110** to fix the multipurpose module **200**. FIG. **2** illustrates the module fixing device **100** having eight locking assemblies **110** arranged on two lines, four locking assemblies **110** for each line.
(32) The locking assembly **110** may be inserted into the fixing hole **210** provided in the multipurpose module **200** and fix the multipurpose module **200**. To this end, the locking assembly **110** and the fixing hole **210** may be disposed to correspond in position to each other. That is, as illustrated in FIGS. **1-3**, the multipurpose module **200** coupled to the module fixing device **100** having the eight locking assemblies **110** arranged on the two lines, four locking assemblies for each line, may also have the fixing holes **210** arranged on two lines, four fixing holes for each line. Therefore, the locking assembly **110** may be inserted into the fixing hole **210**.
(33) Meanwhile, the module fixing device **100** has the guide **130** into which the rail **220** provided on the multipurpose module **200** is inserted, and the rail **220** is inserted into the guide **130**, such that the module fixing device **100** and the multipurpose module **200** may be coupled to each other.
(34) In this case, the locking assembly **110** may be provided in the guide **130** and matched with and inserted into the fixing hole **210** provided in the rail **220**. The structure for coupling the locking assembly **110**, the guide **130**, the fixing hole **210**, and the rail **220** will be described below.
(35) In the embodiment, the module fixing device **100** may be provided on a lower vehicle body **13** in the luggage zone **11**. A user may mount the multipurpose module **200** on the vehicle **10** by aligning and coupling the rail **220** of the multipurpose module **200** to the guide **130** of the module fixing device **100** provided on the lower vehicle body **13** in the luggage zone **11** at the time of coupling the multipurpose module **200** to the vehicle **10**.
(36) FIGS. **3** and **4** are views schematically illustrating the multipurpose module according to the embodiments.
(37) FIGS. **3** and **4** illustrate multipurpose modules **200***a* and **200***b* according to the embodiment. The multipurpose module **200***a* of the embodiment illustrated in FIG. **3** includes a freezer **21**, a refrigerator **22**, and a refrigeration pipe **23**. When the multipurpose module **200***a* is coupled to the vehicle **10**, such that the vehicle **10** may serve as a freezer vehicle.
(38) Meanwhile, the multipurpose module **200***b* of the embodiment illustrated in FIG. **4** includes seats **25** in which passengers are seated. When the multipurpose module **200***b* is coupled to the vehicle **10**, the vehicle **10** may serve as a multi-seat vehicle in which the passengers are seated.
(39) FIGS. **3** and **4** exemplarily illustrate the constituent components provided in the multipurpose module **200**, but the present disclosure is not limited thereto. Various constituent components may be mounted, such that the vehicle **10** may perform various functions.
(40) That is, the multipurpose modules **200** are replaced and mounted according to the purpose of the vehicle **10** according to the embodiment of the present disclosure, such as the transportation of large-scale freight or passengers or the courier service, which makes it possible to ensure various utilization of the vehicle **10**.
(41) Meanwhile, as illustrated in FIGS. **3** and **4**, the multipurpose module **200** may include the fixing hole **210** into which the locking assembly **110** provided and operated in the vehicle **10** is matched and inserted, the rail **220** inserted into the guide **130** provided in the module fixing device **100** of the vehicle **10** and configured to couple the multipurpose module **200** to the vehicle **10**; and a fixing base **230** on which the constituent component is mounted.
(42) As illustrated in FIGS. **3** and **4**, the constituent components, such as the freezer **21**, the refrigerator **22**, the refrigeration pipe **23**, and the seat **25**, are mounted on the fixing base **230** so that the multipurpose module **200** may perform functions according to the purpose. When the purpose is changed, the constituent component is also changed, such that the multipurpose module

200 may perform functions according to the changed purpose.
(43) Meanwhile, the rail **220** may be provided at a lower side of the fixing base **230**. The rail **220** provided at the lower side of the fixing base **230** is inserted into the guide **130** of the module fixing device **100** provided at the lower side of the vehicle **10**, as illustrated in FIG. **2**, such that the multipurpose module **200** may be coupled to the vehicle **10** in a sliding manner.
(44) As illustrated in FIGS. **3** and **4**, the multipurpose module **200** may include the pair of parallel rails **220**. The rails **220** are provided in parallel and coupled, in a sliding manner, to the guides **130** provided in parallel, such that the coupling may be securely maintained.
(45) The coupling between the rail **220** and the guide **130** may allow the multipurpose module **200** to move only in one direction, i.e., the sliding direction relative to the vehicle **10**. In this case, the module fixing device **100** provided in the vehicle **10** may be configured such that the locking assembly **110** is inserted into the fixing hole **210** provided in the multipurpose module **200** to prevent the multipurpose module **200** from sliding.
(46) Meanwhile, the fixing hole **210** may be provided in the rail **220**. More specifically, the rail **220** is a T-shaped rail including a web coupled to the fixing base **230**, and a flange coupled to the web, and the fixing hole **210** may be provided in the web.
(47) The locking assembly **110** may be penetratively inserted into the fixing hole **210** provided in the web to prevent the rail **220** from sliding, such that the multipurpose module **200** may be fixedly mounted on the vehicle.
(48) In the embodiment of the present disclosure, wheel houses **15** for accommodating the road wheels W of the vehicle may protrude in the luggage zone **11**. Therefore, the multipurpose module **200** may have a shape that may not interfere with the wheel house **15** when the multipurpose module **200** is coupled to the vehicle **10** in a sliding manner by the rail **220**.
(49) FIG. **5** is a front view of the module fixing device according to the embodiment, FIG. **6** is a top plan view of the module fixing device according to the embodiment, FIGS. **7** and **8** are cross-sectional front views of the module fixing device according to the embodiment, and FIG. **9** is a view schematically illustrating an operation of the module fixing device according to the embodiment.
(50) FIGS. **5** and **6** illustrate the module fixing device **100** based on the guide **130** at one side (right side) of the module fixing device **100** illustrated in FIG. **2**. In the embodiment of the present disclosure, the module fixing device **100** may include the two or more guides **130** are illustrated in FIG. **2**.
(51) FIG. **5** illustrates a state of the module fixing device **100** when viewed from the rear side of the vehicle **10**. The module fixing device **100** may have the guide **130** into which the rail **220** is inserted. As illustrated in FIG. **5**, the guide **130** may include members provided at two opposite sides **130***a* and **130***b* of a space to define the space into which the rail **220** is inserted.
(52) FIG. **6** is a view illustrating a state of the module fixing device **100** when viewed from the upper side of the vehicle **10**. The module fixing device **100** may include one or more locking assemblies **110**, and the locking controller **120** configured to control the operation of the locking assembly **110**. The locking controller **120** may be electrically connected to the locking assembly **110** and control the operation of the locking controller **120**.
(53) FIG. **7** is a cross-sectional view illustrating a section taken along line A-A in FIG. **6**. As illustrated in FIG. **7**, the module fixing device **100** may have the guide **130** into which the rail **220** is inserted. As illustrated in FIG. **5**, the guide **130** may include members provided at two opposite sides **130***a* and **130***b* of the space to define the space into which the rail **220** is inserted.
(54) FIG. **8** is a cross-sectional view illustrating a section taken along line B-B in FIG. **6**. Referring to FIG. **8**, the structure for mounting the locking assembly **110** and the guide **130** provided in the module fixing device **100** can be ascertained.
(55) Referring to FIG. **8**, the locking assembly **110** according to the embodiment may be provided in the guide **130** and matched with and inserted into the fixing hole **210** provided in the rail **220**.

(56) As illustrated in FIGS. **5** and **8**, the locking assembly **110** may be provided at one side **130***a* of the guide **130** and protrude into the space in the guide **130** when a fixing protrusion **112** operates forward. When the fixing protrusion **112** operates forward, the locking assembly **110** protrudes into the space in the guide **130** and inserted into the fixing hole **210** formed in the rail **220** inserted into the guide **130**, such that the multipurpose module **200** may be fixed to the vehicle **10**.
(57) Meanwhile, a protrusion fixing groove **135** may be provided at the other side **130***b* of the guide **130**, and the fixing protrusion **112** may be inserted into the protrusion fixing groove **135** when the fixing protrusion **112** operates forward. When the fixing protrusion **112** of the locking assembly **110**, which is provided at one side **130***a* of the guide **130**, operates forward, the fixing protrusion **112** may penetrate the fixing hole **210** of the multipurpose module **200** and be inserted into the protrusion fixing groove **135** provided at the other side **130***b* of the guide **130**. As described above, the fixing protrusion **112** provided at one side **130***a* of the guide **130** penetrates the fixing hole **210** and is inserted into the other side **130***b* of the guide **130**, such that the rail **220** and the fixing protrusion **112** may define a cross-locking structure, thereby maintaining secure coupling.
(58) Meanwhile, the module fixing device **100** may further include a support block **140** configured to support the locking assembly **110** to prevent the locking assembly **110** from being moved rearward by a reaction force when the locking assembly **110** operates forward.
(59) Referring to FIG. **8**, the locking assembly **110** according to the embodiment may include: a shape memory spring **111** made of a shape memory alloy; the fixing protrusion **112** connected to the shape memory spring **111** and configured to operate forward to be inserted into the fixing hole **210** or operate rearward to be separated from the fixing hole **210** in accordance with extension or contraction of the shape memory spring **111**, and a heater **113** configured to apply heat to the shape memory spring **111**.
(60) The shape memory spring **111** may operate the fixing protrusion **112** while being extended or contracted by the applied heat. The shape memory spring **111** is made of a shape memory alloy. The shape memory alloy has a feature that is restored to an original shape by a change in crystal structure in accordance with a change in temperature. In the present disclosure, the spring is made of a shape memory alloy and heated, and the restoring force of the shape memory alloy operates the fixing protrusion **112**.
(61) In the embodiment of the present disclosure, in case that no heat is applied to the shape memory spring **111**, the fixing protrusion **112** may operate forward. In case that heat is applied to the shape memory spring **111**, the fixing protrusion **112** may operate rearward.
(62) That is, as illustrated in FIG. **8**, one end of the shape memory spring **111** is coupled to an inner portion of the fixing protrusion **112**, and the other end of the shape memory spring **111** is coupled to a shape memory spring plate **115**. When the shape memory spring **111** is heated by the heater **113**, the shape memory spring **111** is contracted, as illustrated in FIG. **9**, such that the shape memory spring **111** may operate the fixing protrusion **112** rearward in the direction toward the shape memory spring plate **115**. The rearward operation of the fixing protrusion **112** opens the guide **130** to allow the rail **220** to slide.
(63) FIG. **8** illustrates the embodiment in which the fixing protrusion **112** is operated rearward as the shape memory spring **111** is contracted by being heated, but the present disclosure is not limited thereto. In another embodiment, the fixing protrusion **112** may be configured to be operated rearward by the extension of the shape memory spring **111**.
(64) Meanwhile, the locking assembly **110** may further include a reaction force spring **114** configured to apply an elastic force to the fixing protrusion **112** in a forward direction. In case that no heat is applied to the shape memory spring **111**, the reaction force spring **114** operates the fixing protrusion **112** forward, such that the fixing protrusion **112** may be inserted into the fixing hole **210** of the rail **220** coupled to the guide **130**.
(65) Referring to FIG. **8**, the fixing protrusion **112** according to the embodiment includes two cylinders, and the shape memory spring **111** and the reaction force spring **114** are respectively

accommodated in the cylinders. One end of the shape memory spring **111** is coupled to an inner portion of the cylinder of the fixing protrusion **112**, and the other end of the shape memory spring **111** is coupled to the shape memory spring plate **115**. One end of the reaction force spring **114** is coupled to an inner portion of the cylinder of the fixing protrusion **112**, and the other end of the reaction force spring **114** is coupled to a reaction force spring plate **116**. The shape memory spring plate **115** and the reaction force spring plate **116** may be supported by the support block **140**. When the fixing protrusion **112** is operated rearward, the shape memory spring **111** and the reaction force spring **114** are compressed and decreased in length, and the shape memory spring plate **115** and the reaction force spring plate **116** are respectively inserted into the cylinders in which the springs are respectively accommodated, such that the fixing protrusion **112** may be moved rearward toward the support block **140**.
(66) That is, in case that the shape memory spring **111** is deformed in shape as the heat is applied by the heater **113**, the shape memory spring **111** is contracted and operates the fixing protrusion **112** rearward. In case that no heat is applied by the heater **113**, the elastic force of the reaction force spring **114**, which is compressed by the rearward operation, operates the fixing protrusion **112** forward.
(67) The driving power implemented by the shape deformation of the shape memory spring **111** made by heating is higher than the elastic force of the reaction force spring **114**, such that the fixing protrusion **112** may operate rearward. When the shape memory spring **111** is not heated, the elastic force of the shape memory spring **111** is lower than the elastic force of the reaction force spring **114**, such that the fixing protrusion **112** may operate forward.
(68) The driving power implemented by the shape deformation of the shape memory spring **111** and the elastic force of the reaction force spring **114** may be adjusted in accordance with not only the characteristics of the shape memory spring **111** and the reaction force spring **114**, but also lengths of the shape memory spring plate **115** and the reaction force spring plate **116** that are respectively in contact with an end of the shape memory spring **111** and an end of the reaction force spring **114**. That is, a designer of the locking assembly **110** according to the present disclosure may operate the fixing protrusion **112** forward or rearward, in accordance with whether the heat is applied by the heater **113**, by adjusting the characteristics of the shape memory spring **111** and the reaction force spring **114** by changing materials, sizes, and the like of the shape memory spring **111** and the reaction force spring **114** and determining the lengths of the shape memory spring plate **115** and the reaction force spring plate **116**.
(69) In the present disclosure, the state in which the fixing protrusion **112** is operated forward may be maintained when the shape memory spring **111** is not heated by the heater **113**. That is, when the shape memory spring **111** is not heated by the heater **113**, the fixing protrusion **112** is kept inserted into the fixing hole **210** to fix the multipurpose module **200**.
(70) In the embodiment of the present disclosure, the locking controller **120** may control the forward or rearward operation of the fixing protrusion **112** by controlling the heating operation of the heater **113**. At the time of separating the coupled multipurpose module **200**, replacing the multipurpose module **200**, or coupling the multipurpose module **200** to the empty luggage zone **11**, the locking controller **120** may perform control to heat the heater **113** and operate the fixing protrusion **112** rearward to enable the rail **220** of the multipurpose module **200** to slide along the guide **130**.
(71) That is, in the present disclosure, when the locking controller **120** does not perform control, the fixing protrusion **112** operates forward, such that the fixing protrusion **112** may fix the multipurpose module **200** by being fastened to the fixing hole **210** in case that the multipurpose module **200** is coupled, or the fixing protrusion **112** may traverse the guide **130** and be inserted into the protrusion fixing groove **135** provided at the other side in case that the multipurpose module **200** is not coupled.
(72) With the above-mentioned configuration, it is possible to prevent an accident caused by

sudden separation of the multipurpose module **200** because the multipurpose module **200** may be kept coupled without being separated from the vehicle **10** even though the locking controller **120** is broken down.
(73) The heater **113** generates heat and transfer the heat to the shape memory spring **111** under the control of the locking controller **120**. Various heat generator, which generates heat under the control of the locking controller **120**, may be used as the heater **113**. In the embodiment, the heater **113** may be configured as a positive temperature coefficient (PTC) heater.
(74) FIGS. **10** and **11** are views schematically illustrating that the multipurpose module is coupled to the vehicle according to the embodiment.
(75) As illustrated in FIGS. **10** and **11**, the fixing protrusion **112**, which has traversed the guide **130** and moved forward, needs to be operated rearward to couple the multipurpose module **200** to the luggage zone **11** of the vehicle **10**.
(76) Therefore, at the time of coupling the multipurpose module **200**, the multipurpose module **200** needs to be coupled, as illustrated in FIGS. **10** and **11**, by allowing the locking controller **120** to operate the heater **113** first, allowing the heater **113** to heat the shape memory spring **111**, and operating the shape memory spring **111** to operate the fixing protrusion **112** rearward.
(77) As illustrated in FIG. **10**, to couple the multipurpose module **200**, the rail **220** of the multipurpose module **200** is aligned with the guide **130** of the module fixing device **100**, and the rail **220** is coupled to the guide **130** while sliding along the guide **130**, as illustrated in FIG. **11**.
(78) FIGS. **12** and **13** are views schematically illustrating an operation of the module fixing device according to the embodiment.
(79) FIG. **12** is a cross-sectional view illustrating a section taken along line B-B of the module fixing device **100** after the rail **220** of the multipurpose module **200** is inserted into the guide **130** to couple the multipurpose module **200** to the luggage zone **11** of the vehicle **10**, as illustrated in FIGS. **10** and **11**.
(80) The shape memory spring **111** is heated by the heater **113** under the control of the locking controller **120**, such that the fixing protrusion **112** operates forward, and then the rail **220** is inserted into the guide **130**. As illustrated in FIG. **12**, in case that the rail **220** is inserted into the guide **130**, the position of the locking assembly **110** and the position of the fixing hole **210** provided in the rail **220** may be matched and aligned.
(81) As illustrated in FIG. **6**, in case that the plurality of locking assemblies **110** is provided in the guide **130**, the plurality of fixing holes **210** may also be provided in the rail **220** to be inserted into the guide **130**, and the positions of the plurality of fixing holes **210** may correspond to the positions of the plurality of locking assemblies **110**.
(82) Meanwhile, as illustrated in FIG. **13**, the rail **220** according to the embodiment may be the T-shaped rail **220** including the web coupled to the fixing base **230** and the flange coupled to the web, and the guide **130** may have a shape corresponding to the T-shaped rail. In the embodiment in which the rail **220** includes the web and the flange and the guide **130** has the shape corresponding to the rail **220**, the guides **130** may be provided at two opposite sides of the web to prevent the rail **220** from moving to the left and right sides of the guide **130**, and the flange is fastened to the guides **130** to prevent the rail **220** from separating upward from the guide **130**.
(83) After the rail **220** is inserted into the guide **130** and the position of the locking assembly **110** and the position of the fixing hole **210** provided in the rail **220** are matched as illustrated in FIG. **12**, the fixing protrusion **112** of the locking assembly **110** operates forward and is inserted into the fixing hole **210** as illustrated in FIG. **13**.
(84) The fixing protrusion **112** is operated rearward as the shape memory spring **111** is heated by the heater **113**. Thereafter, when the heating operation of the heater **113** is stopped by the locking controller **120**, the driving power implemented by the shape deformation of the shape memory spring **111** is eliminated, and the fixing protrusion **112** is operated forward again by the elastic force of the reaction force spring **114**. The fixing protrusion **112** may be inserted into the fixing hole **210**

by being operated forward by the elastic force of the reaction force spring **114** under the control of the locking controller **120**.
(85) Meanwhile, the fixing protrusion **112** may be not only inserted into the fixing hole **210** while operating forward, but also inserted into the protrusion fixing groove **135** provided at the other side **130***b* of the guide **130** while penetrating the fixing hole **210**. Because the fixing protrusion **112** is provided at one side **130***a* of the guide **130** and inserted into the other side **130***b* of the guide **130** while penetrating the fixing hole **210**, the rail **220** and the fixing protrusion **112** may define a cross-locking structure, thereby maintaining secure coupling.
(86) According to the multipurpose module, the module fixing device, and the locking assembly for a vehicle according to one aspect, it is possible to conveniently change the purposes of the vehicle by using the locking assembly and fixing the multipurpose module equipped with the constituent component suitable for the purpose of the vehicle body.
(87) According to the multipurpose module, the module fixing device, and the locking assembly for a vehicle according to one aspect, it is possible to efficiently couple a coupling object by inserting the locking assembly into the fixing hole provided in the coupling object by operating the locking assembly by using the shape memory spring.
(88) As described above, the embodiments have been described with reference to the accompanying drawings. A person skilled in the art may understand that the present disclosure may be carried out in other forms different from those disclosed in the embodiments without changing the technical spirit or the essential features of the present disclosure. The disclosed embodiments are illustrative and should not be interpreted as being restrictive.

Claims

1. A vehicle comprising: a multipurpose module replaceably coupled to a luggage zone of the vehicle; and a module fixing device coupled to the multipurpose module, and configured to fix the multipurpose module to a vehicle body of the vehicle; wherein the module fixing device comprises: one or more locking assemblies configured to fix the multipurpose module by being matched and inserted into a fixing hole positioned in the multipurpose module; and a locking controller configured to control the locking assembly; wherein the locking assembly comprises: a shape memory spring made of a shape memory alloy; a fixing protrusion connected to the shape memory spring and configured to move forward to be inserted into the fixing hole, or to move rearward to be separated from the fixing hole in accordance with extension or contraction of the shape memory spring; and a heater configured to apply heat to the shape memory spring.
2. The vehicle of claim 1, wherein the shape memory spring operates the fixing protrusion while being extended or contracted by the applied heat.
3. The vehicle of claim 2, wherein the fixing protrusion moves forward when no heat is applied to the shape memory spring, and the fixing protrusion moves rearward when heat is applied to the shape memory spring.
4. The vehicle of claim 3, wherein the locking assembly further comprises a reaction force spring configured to apply an elastic force to the fixing protrusion in a forward direction, and wherein driving power implemented by shape deformation of the shape memory spring is higher than the elastic force of the reaction force spring.
5. The vehicle of claim 2, wherein a rail is provided on the multipurpose module, and a guide is positioned on the module fixing device, such that the module fixing device and the multipurpose module are coupled as the rail is inserted into the guide, and wherein the locking assembly is positioned in the guide and matched and inserted into the fixing hole positioned in the rail.
6. The vehicle of claim 5, wherein the locking assembly is provided at one side of the guide and protrudes when the fixing protrusion moves forward, and wherein a protrusion fixing groove is positioned at the other side of the guide, and the fixing protrusion is inserted into the protrusion

fixing groove when the fixing protrusion moves forward.
7. The vehicle of claim 5, wherein the multipurpose module comprises a fixing base on which a constituent component is mounted, and the rail is positioned at a lower side of the fixing base.
8. The vehicle of claim 7, wherein the rail is a T-shaped rail comprising a web coupled to the fixing base, and a flange coupled to the web, and wherein the fixing hole is positioned in the web.
9. The vehicle of claim 1, wherein the multipurpose module is coupled to the module fixing device in a forward or a rearward direction.
10. A module fixing device for a vehicle, the module fixing device comprising: one or more locking assemblies configured to fix a multipurpose module by being matched and inserted into a fixing hole positioned in the multipurpose module replaceably coupled to a luggage zone of a vehicle; and a locking controller configured to control the locking assembly; wherein the locking assembly comprises: a shape memory spring made of a shape memory alloy; a fixing protrusion connected to the shape memory spring and configured to move forward to be inserted into the fixing hole, or to move rearward to be separated from the fixing hole in accordance with extension or contraction of the shape memory spring; and a heater configured to apply heat to the shape memory spring; wherein the shape memory spring operates the fixing protrusion while being extended or contracted by the applied heat.
11. The module fixing device of claim 10, wherein the fixing protrusion moves forward when no heat is applied to the shape memory spring, and the fixing protrusion moves rearward when heat is applied to the shape memory spring.
12. The module fixing device of claim 11, wherein the locking assembly further comprises a reaction force spring configured to apply an elastic force to the fixing protrusion in a forward direction, and wherein driving power implemented by shape deformation of the shape memory spring is higher than the elastic force of the reaction force spring.
13. The module fixing device of claim 10, wherein the module fixing device has a guide into which a rail positioned on the multipurpose module is inserted, such that the module fixing device and the multipurpose module are coupled as the rail is inserted into the guide, and wherein the locking assembly is provided in the guide and matched and inserted into the fixing hole positioned in the rail.
14. The module fixing device of claim 13, wherein the locking assembly is positioned at one side of the guide and protrudes when the fixing protrusion moves forward, and wherein a protrusion fixing groove is positioned at the other side of the guide, and the fixing protrusion is inserted into the protrusion fixing groove when the fixing protrusion moves forward.
15. A multipurpose module, which is replaceably coupled to a vehicle, the multipurpose module comprising: a fixing hole into which a locking assembly provided and operated in the vehicle is matched and inserted; and a rail inserted into a guide positioned in the vehicle and configured to couple the multipurpose module to the vehicle; wherein the locking assembly comprises: a shape memory spring made of a shape memory alloy; a fixing protrusion connected to the shape memory spring and configured to move forward to be inserted into the fixing hole, or to move rearward to be separated from the fixing hole in accordance with extension or contraction of the shape memory spring; and a heater configured to apply heat to the shape memory spring.
16. The multipurpose module of claim 15, wherein the fixing hole is positioned in the rail.
17. The multipurpose module of claim 15, wherein the multipurpose module further comprises a fixing base on which a constituent component is mounted, and the rail is positioned at a lower side of the fixing base.
18. The multipurpose module of claim 17, wherein the rail is a T-shaped rail comprising a web coupled to the fixing base, and a flange coupled to the web, and wherein the fixing hole is positioned in the web.